



NATIONAL OCCUPATIONAL STANDARD FOR AGRICULTURAL ECONOMIST

NOS.AE.01
First Edition

APPROVING AUTHORITY

This National Occupational Standard has been prepared and published under the authority of the Zambia Qualifications Authority Board on 19th May 2022.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised after every **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Agriculture National Occupational Standards Development Team, upon which the following organisations were represented:

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2. Aquaculture Development Association of Zambia
3. CropLife Zambia
4. Ministry of Fisheries and Livestock (Department of Fisheries)
5. Golden Valley Agriculture Research Trust (GART)
6. Katete College of Agricultural Marketing
7. Ministry of Agriculture (Department of Agriculture)
8. Mulungushi University (MU)
9. Natural Resources Development College (NRDC)
10. University of Zambia (UNZA)
11. Veterinary Council of Zambia
12. Zambia Agriculture Research Institute (ZARI)
13. Zambia National Farmers Union (ZNFU)
14. Zambia Seed Trade Association (ZASTA)
15. Zambia Qualifications Authority (ZAQA)

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to “**provide for the development and implementation of a national qualifications framework; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing**”.

Among other functions, ZAQA is responsible for “**determining national standards for any occupation**”, through various sector specific National Occupational Standards Development Teams (NOSDTs) of experts composed of representation from appropriate authorities, government departments, industry, academia, regulators, consumer associations and non-governmental organizations, etc.

This National Occupational Standard (NOS) has been developed by the Agriculture National Occupational Standards Development Team in accordance with the procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula and learning programmes, in various sectors where the occupation exists. In the Agriculture sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

With the majority of people in Zambia and other countries in Sub-Saharan Africa living in the rural areas, coupled with the ever-increasing pressure on limited finite resources, the analytical skills of Agricultural Economists are as relevant as ever. Analysis that can credibly address the needs of policy makers, non-governmental organizations, agro-business firms, and other institutions requires knowledge and skills of Agricultural Economists. Currently, in this country, the pool of locally based professionals with requisite knowledge, skills and experience is small in relation to the steadily growing demand for their services.

ACRONYMS AND ABBREVIATIONS

AE	Agricultural Economist
CS	Core Skill
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organizational Knowledge
PC	Performance Criteria
PS	Professional Skill
RK	Regulatory Knowledge
RPL	Recognition of Prior Learning
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Function: is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles, which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

1. OVERVIEW

This is an introductory section providing a brief summary and specific information or commentary about the content of the NOS and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.AE.01
Occupation	Agricultural Economist
Job Title	Agricultural Economist
Job Description	Studying and analysing the aspects that influence the agricultural economy and distribution of resources, such as land, raw materials, labour and capital.
Job Purpose	To ensure efficient utilisation of resources through the application of principles and concepts of economics as well as to learn more about the supply and demand of goods and services in the agricultural sector. This includes analysis of production, consumption, and distribution of goods and services.
ZQF Level	7
Sector	Agriculture
Sub sector	Agriculture Economics
Other Economic Sector(s) in which the Occupation is Practiced	Education (Academia); Financial and Insurance sector; Commercial and Manufacturing sector; Various governmental ministries; and self-employment
Other Similar Jobs that can be performed by an Agricultural Economist	Monitoring and Evaluation Officer; Project Managers; Bankers
Minimum Educational Job Entry Qualification(s)	Bachelor's Degree in Agricultural Economics
Practicing License Requirements (if any)	None
Training/RPL	A deep interest in agriculture and the natural environment as well as in economics. Prior knowledge of mathematics, biology, physics and chemistry is required.
Minimum Job Entry Age	21 years
Prior Experience	None – Field practicals and attachments will suffice.
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competences that an Agricultural Economist must possess to be successful in his/her job role. It is applicable to Agricultural Economists working in public or private organisations or self-employed.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess:

- Creativity
- Problem solving skills
- Analytical skills
- Mathematics skills
- Integrity and respect for confidentiality
- Interpersonal skills
- Commercial Awareness
- Attention to details
- Ability to communicate effectively and clearly
- Self-motivated and team worker
- Ability to plan and prioritize,
- Quality consciousness

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 6 units representing the tasks that a job holder should undertake in his/her day to day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This unit is about fundamentals in economics, soils, crops, and livestock production].

Unit No.	01
Unit Title	Fundamentals in economics, soils, plants and animal production
Description	This unit is about understanding the fundamental principles of economics, soils, plants and animal production.
Scope	This unit covers the following: • Modern simple treatment of micro and macro-economic theory measurement and policy • Genesis and composition of soil and the scientific basis for promoting good husbandry of soil • Plant functions • Forms and structures, and functions of the forms and parts of farm animals
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Modern simple treatment of micro and macro-economic theory measurement and policy	To be competent, the individual must be able to: PC1. Articulate the concepts of supply, demand, cardinal utility and ordinal utility PC2. Apply the concepts of elasticity and production PC3. Measure gross national product PC4. Measure inflation and employment rates PC5. Measure investment interest rate and exchange rate PC6. Analyse fiscal policy, monetary policy and international trade
Genesis and composition of soil and the scientific basis for promoting good husbandry of soil	To be competent, the individual must be able to: PC7. Synthesize the information carried in the soil profile and vegetation PC8. Interpret the soil developmental processes PC9. Make broad informed evaluations of the capability of soil and suitability for different and varied uses PC10. Conduct relevant soil chemical, physical and biological analysis PC11. Interpret soil analytical data for application to practical situations in the field
Plant functions	To be competent, the individual must be able to: PC12. Explain how the plant functions PC13. Relate the different phases of plant development PC14. Relate the stresses a plant can undergo during development PC15. Classify crop plants

Forms and structures, and functions of the forms and parts of farm animals	To be competent, the individual must be able to: PC16. Articulate gross anatomical parts of farm animals PC17. Differentiate the micro anatomy of the various parts of an animal PC18. Relate the different anatomical parts to the functions they perform PC19. Apply physiology principles in real life situations
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company regulations and global best practices OK2. Guidelines for liaising with industry players OK3. Code of business conduct OK4. Job responsibilities/duties
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Basic agricultural production practices TK2. Economic concepts measurement TK3. The morphological (botanical) features of crop plants TK4. Principles of resource capture and metabolism within plants. TK5. Principles of crop production based on knowledge of plant classification and plant physiology TK6. Terminologies used in physiology TK7. Animal body function at cellular level
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory requirements in the agricultural sector
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, memos, other company documents. CS4. Read from different sources- books, screens in machines and signage.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS5. Express statements or information clearly so that others can hear and understand.

	CS6. Participate in and understand the main points of simple discussions. CS7. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 2 [This unit is about the Agricultural Production Economy].

Unit No.	02
Unit Title	Agricultural Production Economy
Description	This unit is about knowing various agricultural production enterprises, and analysing the relationship between the inputs including labour, and the outputs.
Scope	This unit covers the following: • Field crops production • Livestock (pig, ruminant, poultry) and aquaculture production • Intermediate micro and macro-economic theory measurement and policy • Relationships between inputs (labour, capital, and land) and outputs • Relations of people who live in the rural areas • Transfer of scientific research and new knowledge to agricultural practices through farmer education
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Field crops production	To be competent, the individual must be able to: PC1. Articulate production requirements of major field crops and under-utilized crops
Livestock (pig, ruminant, poultry) and aquaculture production	To be competent, the individual must be able to: PC2. Relate the state and limitations of livestock and aquaculture industries PC3. Perform management operations involved in livestock and aquaculture husbandry.
Intermediate micro and macro-economic theory measurement and policy	To be competent, the individual must be able to: PC4. Apply mathematical tools such as calculus for use in microeconomic analysis PC5. Analyze consumer and producer behaviour through tools such as constrained optimisation, comparative statics and equilibrium analysis PC6. Analyze decision making in different market structures PC7. Apply general equilibrium analysis to understand interactions in several markets affecting economic agents PC8. Analyze externalities and public goods and how they affect economic efficiency PC9. Apply macroeconomic theory to the study of field courses within the discipline of agricultural economics
Relationships between inputs (labour, capital, and land) and outputs	To be competent, the individual must be able to: PC10. Apply the key concepts of production economics in an agribusiness setup PC11. Demonstrate the use and application of mathematics in solving production economics problems in an agribusiness

	PC12. Demonstrate the use and application of graphical presentations in solving production economics problems in an agribusiness
Relations of people who live in the rural areas	To be competent, the individual must be able to: PC13. Distinguish between rural and urban ways of life PC14. Examine relevant tools and techniques of analyzing societies PC15. Explain the rural development strategies used in Developing countries
Transfer of scientific research and new knowledge to agricultural practices through farmer education	To be competent, the individual must be able to: PC16. Apply current trends in implementing agricultural extension
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company regulations and global best practices OK2. Guidelines for liaising with industry players OK3. Code of business conduct OK4. Job responsibilities/duties
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Agricultural production and economics TK2. Essential aspects of macroeconomics, such as national TK3. Income determination, fiscal policy, monetary policy, inflation and stabilisation policy. TK4. Assumptions about the nature and objectives of agricultural extension TK5. Differences in the organizations and practices of agricultural extension
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory requirements in the agricultural sector
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English and give simple concise instructions.
	Reading Skills
	The individual on the job must be able to:

	CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, memos, other company documents. CS4. Read from different sources- books, screens in machines and signage. CS5. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS6. Express statements or information clearly so that others can hear and understand. CS7. Participate in and understand the main points of simple discussions. CS8. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 3 [This unit is about Farm and Financial Management].

Unit No.	03
Unit Title	Farm and Financial Management
Description	This unit is about understanding aspects of the farm management process itself, agricultural planning and the principles of financing
Scope	This unit covers the following: • Agricultural organization and administration • Agricultural project planning and appraisal • Project monitoring and evaluation • Farm management • Agricultural Finance
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Agricultural organisation and administration	To be competent, the individual must be able to: PC1. Apply the five functions of management, that is, planning, organizing, directing, staffing and controlling PC2. Apply management theory and concepts to management practice
Agricultural project planning and appraisal	To be competent, the individual must be able to: PC3. Develop agricultural projects PC4. Appraise agricultural projects PC5. Apply techniques (or investment criteria) for choosing among competing projects PC6. Apply knowledge and skills necessary for optimally allocating resources
Project monitoring and evaluation	To be competent, the individual must be able to: PC7. Apply the key concepts of project monitoring and evaluation in any project setup. PC8. Demonstrate their ability to develop a monitoring and evaluation system
Farm management	To be competent, the individual must be able to: PC9. Apply economic principles to determine conditions that maximize profits in the choice of production levels, input and outputs combinations PC10. Develop plans and budgeting techniques that aid in successful decision making on the farm PC11. Analyze the financial performance of a farm business PC12. Evaluate risk and uncertainty in decision making
Agricultural finance	To be competent, the individual must be able to: PC13. Examine the various options available for financing agriculture development PC14. Determine the financial viability of agricultural projects/programmes

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company regulations and global best practices OK2. Guidelines for liaising with industry players OK3. Code of business conduct OK4. Job responsibilities/duties
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Agricultural organisation and administration TK2. Project planning, appraisal, monitoring and evaluation TK3. Agricultural farm management and finance TK4. Importance of agricultural finance in an economy TK5. The main challenges facing agriculture finance in developing countries
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory requirements in the agricultural sector
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, prepare and submit comprehensive and consolidated reports.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, memos, other company documents. CS4. Read from different sources- books, screens in machines and signage.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS5. Express statements or information clearly so that others can hear and understand. CS6. Participate in and understand the main points of simple discussions. CS7. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.

	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with the intent of satisfying their requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7 Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 4 [This unit is about Agricultural Entrepreneurship and Marketing Environment].

Unit No.	04
Unit Title	Agricultural Entrepreneurship and Marketing Environment
Description	This unit is about entrepreneurial agribusinesses as well as understanding all aspects of marketing such as the price system and market types.
Scope	This unit covers the following: • Agricultural marketing and pricing • International agricultural markets, trade and development • Basics of sustainable agri-entrepreneurship • Agricultural entrepreneurship development • Agribusiness management
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Agricultural marketing and pricing	To be competent, the individual must be able to: PC1. Analyse marketing institutions and commodity markets PC2. Conduct empirical price analysis
International agricultural markets, trade and development	To be competent, the individual must be able to: PC3. Explain why nations trade; PC4. Examine trade barriers and the benefits of free trade to the overall country; PC5. Apply international trade and finance theory to the domestic policy environment.

Basics of sustainable agri-entrepreneurship	To be competent, the individual must be able to: PC6. Identify the varied opportunities that exist in the agriculture sector and entire agribusiness system PC7. Demonstrate how they can leverage the bio economy and circular economy for transformative agribusiness ideas PC8. Explain the significance of entrepreneurship, particularly the small agribusinesses to the national and global economy PC9. Explain the importance of ethics for entrepreneurs. PC10. Critically evaluate the effect of external environmental factors on agribusiness survival and growth
Agribusiness entrepreneurship development	To be competent, the individual must be able to: PC11. Identify and develop a transformative business idea PC12. Assess the importance of customer profiling when developing a new product for the market PC13. Conduct a market and futures research for business idea development
Agribusiness management	To be competent, the individual must be able to: PC14. Evaluate the typical marketing issues faced by small agribusinesses PC15. Evaluate the marketing strategies and methods available to an entrepreneur PC16. Demonstrate the importance of reporting of financial statements in sound financial management PC17. Identify key financial issues faced by the entrepreneurial small agribusinesses PC18. Evaluate ways in which an agribusiness creates value and how this can be realized PC19. Assess the succession issues faced by agribusinesses owners in a variety of contexts including family businesses
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company regulations and global best practices OK2. Guidelines for liaising with industry players OK3. Code of business conduct OK4. Job responsibilities/duties
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Marketing and entrepreneurship skills TK2. Price determination concepts for agricultural commodities TK3. Factors that are critical to price variability of agricultural products TK4. The nature of and contexts of the different types entrepreneurship TK5. The different stages of business development

C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory requirements in the agricultural sector
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, prepare and submit comprehensive and consolidated reports.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, memos, other company documents. CS4. Read from different sources- books, screens in machines and signage.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS5. Express statements or information clearly so that others can hear and understand. CS6. Participate in and understand the main points of simple discussions. CS7. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks

	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 5 [This unit is about Sustainable Agricultural Development and Policy Analysis].

Unit No.	05
Unit Title	Sustainable agricultural development and policy analysis
Description	This unit is about understanding the interaction between agriculture and other sectors including the environment, and the policy design and analysis domains.
Scope	This unit covers the following: - Interactions of agricultural production process and the environment - Agricultural policy design and analysis
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Interactions of agricultural production process and the environment	To be competent, the individual must be able to: PC1. Analyse environmental problems such as air and water pollution, deforestation, soil degradation and sedimentation. PC2. Identify specific environmental problems associated with agriculture production PC3. Employ the approach of environmental economics to the solution of environmental problems
Agricultural policy design and analysis	To be competent, the individual must be able to: PC4. Design an agricultural policy PC5. Recognize the alternative policy instruments for government intervention PC6. Apply various tools and approaches of agricultural policy analysis PC7. Assess the efficiency of government institutions and the private sector in agricultural research
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company regulations and global best practices OK2. Guidelines for liaising with industry players OK3. Code of business conduct OK4. Job responsibilities/duties

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Goals and objectives of agricultural policy TK2. Traditional policies of food security TK3. Economic and environmental models of policy analysis
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory requirements in the agricultural sector
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, prepare and submit comprehensive and consolidated reports.
	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, memos, other company documents. CS4. Read from different sources- books, screens in machines and signage.
	Oral Communication (Listening and Speaking skills)
B. Professional Skills	The individual on the job must be able to : CS5. Express statements or information clearly so that others can hear and understand. CS6. Participate in and understand the main points of simple discussions. CS7. Respond appropriately to any queries.
	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities

	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

UNIT 6 [This unit is about agricultural economics operational research].

Unit No.	06
Unit Title	Agricultural Economics Operational Research
Description	This unit is about understanding various tools and the application of economic simulation and optimisation techniques on agricultural problems. This is basically a support unit for the above units.
Scope	This unit covers the following: • Mathematics for economics • Statistics for agriculture • Quantitative methods in agricultural economics • Econometrics • Research Methodology • Communication • Research Projects
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Mathematics for economics	To be competent, the individual must be able to: PC1. Apply mathematical concepts in the rigorous statement of economic problems PC2. Employ mathematical techniques to solve optimization problems in economics PC3. Employ the techniques of optimization and programming
Statistics for agriculture	To be competent, the individual must be able to: PC4. Use statistics in order to gauge the probability of a particular outcome
Quantitative methods in agricultural economics	To be competent, the individual must be able to: PC5. Implement appropriate techniques in resource management PC6. Apply linear programming to an array of agricultural decision problems

Econometrics	To be competent, the individual must be able to: PC7. To use econometric techniques in empirical economic analysis PC8. Understand when to adopt a particular model or technique/estimator PC9. Implement econometric techniques in appropriate computer packages PC10. Interpret econometric results
Research methodology	To be competent, the individual must be able to: PC11. Identify researchable topics and research problems in agricultural economics and agricultural extension PC12. Develop a research proposal PC13. Apply the scientific research process to real life research problems
Communication	To be competent, the individual must be able to: PC14. Apply communication concepts and skills PC15. Apply key concepts of professional and scientific writing
Research Projects	To be competent, the individual must be able to: PC16. Develop and present a research proposal PC17. Implement and write a research project PC18. Orally present their research project
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company regulations and global best practices OK2. Guidelines for liaising with industry players OK3. Code of business conduct OK4. Job responsibilities/duties
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Set theory, elementary differential and integral calculus and matrix algebra TK2. Various econometric analytical models TK3. The nature, scope and methodology of econometrics
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Regulatory requirements in the agricultural sector
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English, prepare and submit comprehensive and consolidated reports.
	Reading Skills

	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, memos, other company documents. CS4. Read from different sources- books, screens in machines and signage.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS5. Express statements or information clearly so that others can hear and understand. CS6. Participate in and understand the main points of simple discussions. CS7. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow the code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisors, and co-workers to perform day-to-day activities. PS13. Ask relevant questions for better understanding.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to; Desk Computers/Laptop, Printer and relevant Software programs, Personal protective equipment (PPE).

6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER

Dilemmas associated with the job of an Agricultural Economist include long working hours, time pressure to complete tasks and from a diverse multi-stakeholders, working in extreme weather such as hot and cold conditions, working in noisy environments etc.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include, ensuring good time management and planning, participating in workplace safety sensitization and awareness, supporting capacity building through training, managing work stress, adhering to company's safety and standard operating procedures at all times, paying attention to detail, consulting extensively within and outside one's department/team.

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include working in the field under varied weather conditions, in the office for long hours, working areas that are sometimes noisy.

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organization

Parties involved/interacting with the job holder who are internal to the organization include supervisors, subordinates, and other employees.

8.2 External/Outside the Organization

Parties involved/interacting with the job holder who are external include; non-governmental organisations and donor agencies on matters of programme planning and development, customers/clients, government regulators, trainers, suppliers of equipment/tools/consumables, academia etc.

9. PHYSICAL DEMANDS ON THE BODY

- Minimum physical effort for both office and field work.
- Be able to walk and stand for some periods of time;
- Substantial mental effort associated with preparation of technical reports;
- Be able to use fingers, hands and feet with ease to complete the assigned task (dexterity);
- Etc.

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

- A.1.1** Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

- A.1.2** Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.AE.01		
ZQF Level	7	Version Number	01
Sector	Agriculture	Date of Approval	19 th May 2022
Sub Sector		Date of Last Review	N/A
Occupation	Agriculture Economics	Date of Next Review	May 2027

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